BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Mixed Paper 1 and Paper 2 Retrieval

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. A GBP 5-per-ticket program uses total = tickets + 5. What is the correction?

2. What does an IDE's run-time environment let you do?

A) Repair physical CPU faults
B) Rearrange HDD fragments
C) Manage network user accounts
D) Run and test code

3. "An IPv4 address is always six pairs of hexadecimal digits separated by colons." Correct this. Explain why.

4. An IDE finds a missing bracket. Once fixed, the program runs but gives the wrong total. Why did the IDE not prove the logic correct? What should you do next?

Today's objectives

Explain the core ideas of mixed paper 1 and paper 2 retrieval using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Synoptic retrieval
Recall previously learned Paper 1 and Paper 2 material in mixed order.

Choose the method
Decide whether a task needs an explanation, calculation, trace or evaluation.

Link systems and coding
Variables, memory and data representation affect the way programs execute.

Identify gaps
Use wrong answers to choose the next targeted practice rather than repeat every topic.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Synoptic retrieval	Recall previously learned Paper 1 and Paper 2 material in mixed order.
Choose the method	Decide whether a task needs an explanation, calculation, trace or evaluation.
Link systems and coding	Variables, memory and data representation affect the way programs execute.
Identify gaps	Use wrong answers to choose the next targeted practice rather than repeat every topic.

I Do - One worked example

A monochrome 100 x 50 bitmap uses 1 bit/pixel. Calculate its uncompressed file size in bytes.

1. Compute pixels: 100 x 50=5000.
2. Multiply by one bit per pixel: 5000 bits.
3. Convert 5000 / 8=625 bytes.
4. State assumptions: ignoring metadata and compression.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A mono sound clip lasts 5 seconds at 8000 samples/s and 8 bits/sample. Find bytes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the MOD calculation that finds leftover bits.

```
bits = 203
leftover = ...
print(leftover)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Calculate the denary value of unsigned binary 10101010. [3 marks]

Q3. Explain why binary search requires a sorted list. [3 marks]

Q4. Explain one advantage and one limitation of virtual memory. [4 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Image file size in bytes equals width x height x bit depth without any conversion. ☐ T ☐ F
2. Recall previously learned Paper 1 and Paper 2 material in mixed order. ☐ T ☐ F
3. Decide whether a task needs an explanation, calculation, trace or evaluation. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Use total = tickets * 5; addition does not calculate GBP 5 for each ticket. ☐

Q2 D - Run and test code ☐

Q3 IPv4 supports addressing and routing. It is usually written as four decimal numbers separated by dots. ☐

Q4 The IDE found a syntax error, not a logic error. Trace the calculation. Test inputs with known totals and compare results. ☐

Code First - Model

```
bits = 203
leftover = bits MOD 8
print(leftover)
```

Output: 3

We Do - Check the model

1. 40,000 bytes. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 128+32+8+2 = 170. ☐

Q3 Midpoint comparisons only allow safe elimination of half the list when values are ordered. ☐

Q4 Lets the OS accommodate memory demand beyond physical RAM; uses slower secondary storage, so paging can reduce performance. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: